Mathematics in CP

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Sections

- Series and sums
- Ceiling and Floor functions
- Divisors and divisibility
- Prime Numbers, detection and generation
- Modular arithmetic
- Binary Exponentiation
- Extended Euclidean Algorithm

Sums and Series

- https://en.wikipedia.org/wiki/Summation#General_identities
- Series formulas

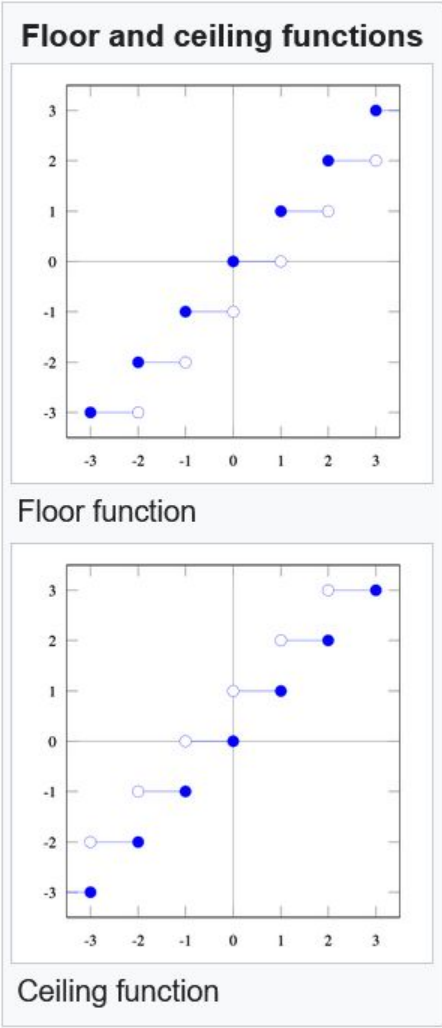
https://codeforces.com/contest/1853/problem/B

Ceiling and Floor Functions

- ceil(x): Smallest integer >= x
- floor(x): Largest integer <= x
- x - 1 < floor(x) <= x <= ceil(x) < x + 1
  - Because in a half open interval of length 1, there is exactly 1 integer.
- ceil(x) = - floor(-x)
- When m is positive:

$$\left\lceil \frac{n}{m} \right\rceil = \left\lfloor \frac{n+m-1}{m} \right\rfloor = \left\lfloor \frac{n-1}{m} \right\rfloor + 1,$$

- Remainder $$x \bmod y = x - y \left\lfloor \frac{x}{y} \right\rfloor$$



https://codeforces.com/problemset/problem/616/E

Divisors and divisibility

- If a number D divides X (leaves remainder = 0), we write:

  D | X (read as “D divides X”)

  6 | 12
- D | X also means, that there exists an integer k such that D * k = X.
- If D | X, then D is a divisor/factor of X. And X is a multiple of D.
- Often, we may need to find all the divisors of a number.

  12 = 1 * 12 = 2 * 6 = 3 * 4
- Divisors come in pairs. If d | x, then (x/d) | x too.
- How to find all divisors of a number quickly?
- How to find all divisors of all the numbers from 1 to n fast?

```
vector<vector<int>> divisors;

int main () {
    int n; cin >> n;
    divisors.resize(n + 1);

    for (int i = 1; i <= n; ++i)
        for (int j = i; j <= n; j += i)
            divisors[j].push_back(i);

}
```

Divisibility properties

- https://math.libretexts.org/Bookshelves/Combinatorics_and_Discrete_Mathematics/Elementary_Number_Theory_(Barrus_and_Clark)/01%3A_Chapters/1.04%3A_Elementary_Divisibility_Properties

Prime Numbers

- A positive integer p which is only divisible by 1 and p is called a prime.
- Why care about primes? Fundamental Theorem of Arithmetic

Primality Testing

- How to check if a given number is a prime?
-

```
bool isPrime (ll x) {
    if (x < 2) return false;
    for (ll i = 2; i * i <= x; ++i)
        if (x % i == 0)
            return false;
    return true;
}
```

Prime Generation

- How to generate all prime numbers upto n?
- Sieve of Eratosthenes

```
vector<bool> isc; // is composite
vector<int> pr;   // final prime list

void sieve (int n) {
    for (int i = 2; i * 1LL * i <= n; ++i) if (!isc[i]) {
        for (int j = i * i; j <= n; j += i)
            isc[j] = true;
    }
    for (int i = 2; i <= n; ++i) if (!isc[i])
        pr.push_back(i);
}
```

- Complexity: O(n ln ln n)